

Assignment 2

a) Write a program using fork to create a child process. The parent process should sort elements in ascending order and child process should sort elements in descending order.

```
//unnatti gogna
//assign2.a
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>
void bubbleSortAsc(int arr[], int n){
    for(int i = 0; i < n-1; i++){
        for(int j = 0; j < n-i-1; j++){
            if(arr[j] > arr[j+1]){
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

void bubbleSortDesc(int arr[], int n){
    for(int i = 0; i < n-1; i++){
        for(int j = 0; j < n-i-1; j++){
            if(arr[j] < arr[j+1]){
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

void display(int arr[], int n){
    for(int i = 0; i < n; i++){
        printf("%d\t", arr[i]);
    }
    printf("\n");
}

int main(){
    printf("Enter the number of elements: ");
```

```

int n;
scanf("%d", &n);

int elements[n];
for(int i=0; i<n; i++){
    printf("Enter element %d: ", i+1);
    scanf("%d", &elements[i]);
}

pid_t pid = fork();

if(pid > 0){
    wait(NULL);
    bubbleSortAsc(elements, n);
    printf("Ascending order: ");
    display(elements, n);
}
else if(pid == 0){
    bubbleSortDesc(elements, n);
    printf("Descending order: ");
    display(elements, n);
    exit(0);
}
else{
    printf("Fork failed.\n");
}
return 0;
}

```

/*OUTPUT

```

Enter the number of elements: 3
Enter element 1: 5
Enter element 2: 3
Enter element 3: 1
Descending order: 5 3 1
Ascending order: 1 3 5

```

*/

b) Write a program using fork to create a child process. Then, make that child process as Orphan Process

```
//unnatti gogna
//assign2.b
#include<stdio.h>
#include<sys/types.h>
#include<unistd.h>
#include<stdlib.h>
int main() {
pid_t pid;
pid=fork();

if(pid>0){
printf("this is parent process with id: %d\n",
getppid());
printf("this is child process with id: %d\n", getpid());
}
else{
if(pid==0){
sleep(10);
printf("this is child process with init process id:
%d\n", getppid());
printf("the child process id is: %d\n", getpid());
}
}
return 0;
}

/*output*/
/*
sdc-user@VY-7F-IOSTC-023 ~ % gcc orphanProcess.c
sdc-user@VY-7F-IOSTC-023 ~ % ./a.out
this is parent process with id: 30599
this is child process with id: 32317
sdc-user@VY-7F-IOSTC-023 ~ % this is child process with
init process id: 1
the child process id is: 32318
*/
```

c) Write a program using fork to create a child process. Then, make that child process as Zombie process

```
//unnatti gogna
//assign2.c
#include<stdio.h>
#include<sys/types.h>
#include<unistd.h>
#include<stdlib.h>
int main()
{
    pid_t pid;
    if((pid=fork())<0){
        printf("\tfork error\n");
    }
    else{
        if(pid==0){
            printf("child process id is: %d", getpid());
        }
        else{
            sleep(10);
            printf("****parent\n");
            system("ps -axj | tail");
        }
    }
    exit(0);
}
```

```
/*output*/
/*sdc-user@VY-7F-IOSTC-023 ~ % gcc zombieProcess.c
sdc-user@VY-7F-IOSTC-023 ~ % ./a.out
child process id is: 31479****parent
sdc-user          31461      1 31461          0      1 S      ??
0:00.07 /System/Library/Frameworks/
CoreServices.framework/Frameworks/Metadata.framework/
Versions/A/Support/mdworker_shared -s mdworker -c
MDSImporterWorker -m com.apple.mdworker.shared
sdc-user          31462      1 31462          0      1 S      ??
0:00.08 /System/Library/Frameworks/
CoreServices.framework/Frameworks/Metadata.framework/
Versions/A/Support/mdworker_shared -s mdworker -c
MDSImporterWorker -m com.apple.mdworker.shared
root              108        1   108          0      0 Ss+    ??
0:00.33 /usr/libexec/remoted
```

root	30598	30597	30598	0	0	Ss	s000
0:00.01 login -pf sdc-user							
sdc-user	30599	30598	30599	0	1	S	s000
0:00.11 -zsh							
sdc-user	31478	30599	31478	0	1	S+	s000
0:00.01 ./a.out							
sdc-user	31479	31478	31478	0	1	Z+	s000
0:00.00 <defunct>							
sdc-user	31487	31478	31478	0	1	S+	s000
0:00.01 sh -c ps -axj tail							
root	31488	31487	31478	0	1	R+	s000
0:00.00 ps -axj							
sdc-user	31489	31487	31478	0	1	S+	s000
0:00.00 tail							
sdc-user@VY-7F-IOSTC-023 ~ % */							